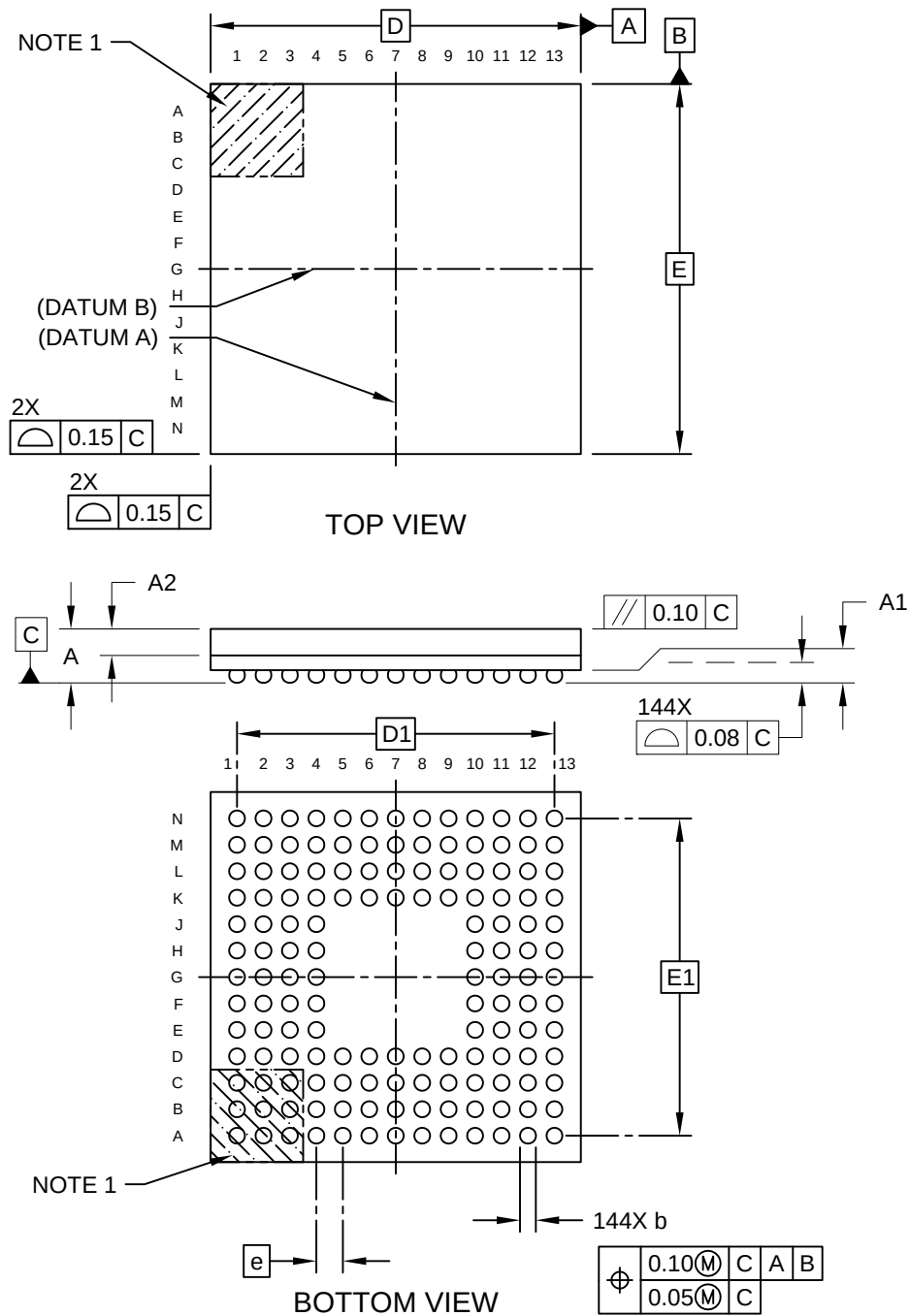


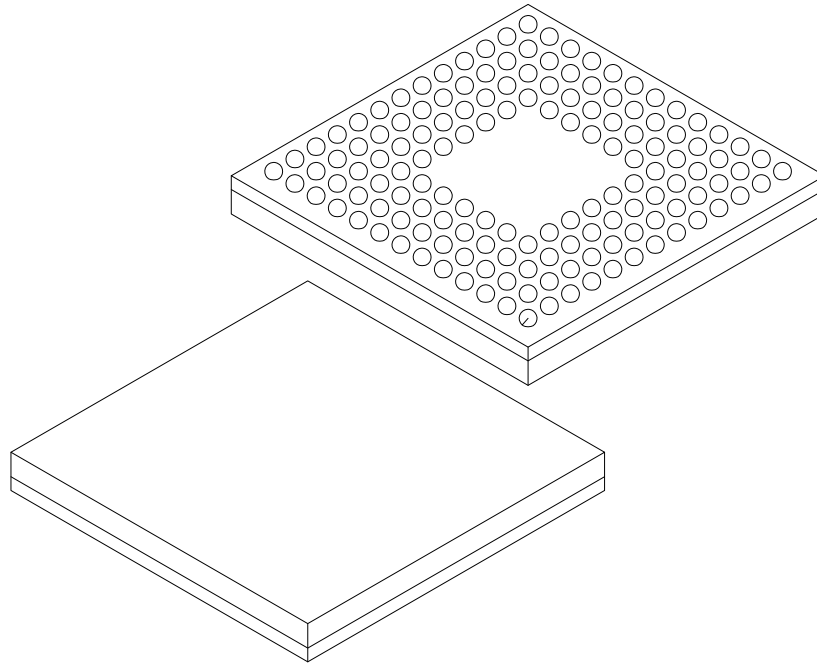
144-Ball Thin Fine Pitch Ball Grid Array [JWX] - 7x7 mm Body (TFBGA)

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	144		
Pitch	e	0.50 BSC		
Overall Height	A	-	-	1.02
Standoff	A1	0.15	0.24	-
Molded Cap Height	A2	0.45	0.50	0.55
Overall Length	D	7.00 BSC		
Overall Pitch	D1	6.00 BSC		
Overall Width	E	7.00 BSC		
Overall Pitch	D1	6.00 BSC		
Ball Diameter	b	0.25	0.30	0.35

Notes:

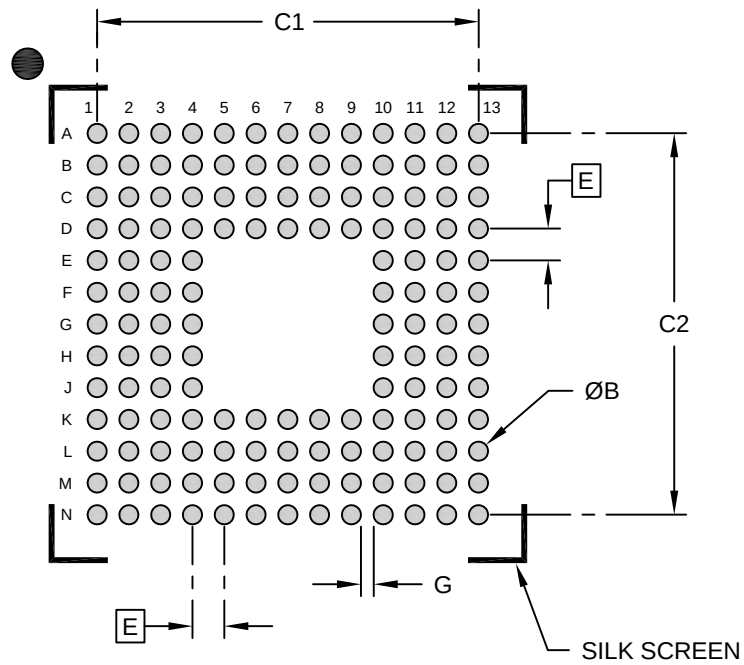
1. Terminal A1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

REF: Reference Dimension, usually without tolerance, for information purposes only.

144-Ball Thin Fine Pitch Ball Grid Array [JWX] - 7x7 mm Body (TFBGA)

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Overall Contact Pitch	C1		6.00	
Overall Contact Pitch	C2		6.00	
Contact Diameter (X 144)	X1			0.30
Spacing Between Contacts	G	0.30		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.